

Bradley Heim

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EDUCATION

Purdue University

B.S. in Aeronautical & Astronautical Engineering, College of Engineering

B.S. in Integrated Business and Engineering, Mitch Daniels School of Business

Specialization/Concentrations: **Systems Design** | **Data Analytics** | **Leadership**

West Lafayette, IN

May 2027

3.73/4.0 GPA

SKILLS & INTERESTS

Engineering: Siemens NX, ANSYS, Fusion 360, CAM, CNC Machining, Waterjet, Laser Cutting, Additive Manufacturing

Systems Engineering: Requirements Development, Requirements Traceability, Verification & Validation Planning, Interface Definition, Trade Studies, Risk Analysis, System Architecture, Technical Writing, Project Management

Programming & Interest: Python, MATLAB, R, Java, Excel, Skiing, Traveling, Running, Space Exploration, Volleyball

PROFESSIONAL EXPERIENCE

Voyager Technologies

Cleveland, OH

Growth & Business Development Intern

June 2026 – August 2026

- Developed three NASA proposal responses from concept through submission, translating mission concepts and solicitation requirements into CONOPS and technical requirements analyses with technical subject-matter experts
- Built a modular Python/HTML AI tool integrating SAM.gov APIs, web scraping, and Anthropic LLMs to evaluate hundreds of NASA opportunities and generate traceable capability-gap analyses, reducing initial screening time from ~30 to ~5 minutes
- Reconciled a \$12M+ BOM spanning 181 unique parts and tens of thousands of components, identifying \$869K in projected savings through data validation and updated supplier pricing

TECHNICAL EXPERIENCE

Space Biomedical Systems / Purdue Vertically Integrated Projects

West Lafayette, IN

Founder & President | *Project Lead, Microgravity Blood Transfusion System*

August 2025 – Present

- Founded Space Biomedical Systems and lead a 10-person multidisciplinary team developing a microgravity blood transfusion system, directing system architecture, requirements development and traceability, interface definition, risk analysis, verification planning, and subsystem integration across fluid, mechanical, electrical, controls, and HMI.
- Designed and fabricated peristaltic pump and air-management prototypes using Siemens NX, motor sizing, CNC machining, waterjetting, and additive manufacturing while advancing the system toward integrated blood rehydration, fluid delivery, sensing, embedded controls, and user interface functions.
- Selected as 1 of 12 finalist teams in NASA's 2026 Human Lander Challenge, earning \$9,000 in funding and presenting a 25-minute technical briefing, 20-minute Q&A, and poster to NASA subject-matter experts.

Purdue Space Program - Satellites

West Lafayette, IN

Structures, Mechanisms and Thermals

August 2024 – May 2026

- Supported structures and thermals for a 3U CubeSat designed to the Cal Poly CubeSat Design Specification, contributing to manufacturing and implementing a vacuum-chuck fixture for CNC machining of satellite assemblies
- Applied ANSYS to spacecraft thermal analysis while advancing the vehicle toward Preliminary Design Review following successful System Design Review

Aerial Firefighting Aircraft Design Competition

West Lafayette, IN

1st place team / *Northrop Grumman S.P.A.C.E. Award*

January 2025 - May 2025

- Designed an unmanned tanker aircraft for wildfire retardant delivery, applying systems engineering practices including risk analysis, aerodynamic sizing, propulsion selection, and mission performance analysis in MATLAB, co-authoring a 58-page technical design report and recorded video presentation for competition judges
- Awarded 1st Place across all classes in Purdue's Northrop Grumman S.P.A.C.E. Award aircraft design competition

IBE Business Capstone

West Lafayette, IN

External Consultant

January 2026 – April 2026

- Evaluated AI/ML approaches to supply-chain risk management for a semiconductor equipment manufacturer, analyzing disruption scenarios and geopolitical risks and presenting recommendations to 6 company stakeholders

JOB EXPERIENCE

Purdue Football Student Video Manager

March 2024 – Present